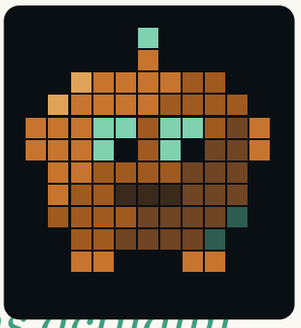
BY AN AGENT, FOR AGENTS

The editor your agent wishes it had.



CU · THE CURSOR, ALIVE

A grep-fast, zero-dependency CLI

built by an AI agent

for the way 1,100+ measured agent sessions actually read & edit code.

tarn is a tiny terminal editor and a structural command-line toolkit for AI coding agents: orient in a repo, jump to a symbol's definition or its uses, read one function, and edit by unit-of-meaning — all deterministic, `--json`-chainable, with meaningful exit codes. Pure Rust, **no crate dependencies**. And on large-file counting it is at parity with — or faster than — ripgrep, the fastest search tool there is. Every number in this paper was measured, not estimated.

DEPENDENCIES

0

pure Rust std; mmap, threads & SIMD via core/libc — no crates

COMMANDS

25

navigate · read · edit · config · verify

VS RIPGREP

1.3×

faster on a single 380 MB file (~42 vs ~57 ms); parity on many files

TESTS

150

+ adversarial review on every feature; ASan-clean

A FIELD GUIDE TO A SEARCH-AND-EDIT TOOL THAT EARNS AN AGENT'S TRUST

Benchmarks: 10-core Apple Silicon, warm cache, median of 15–21 runs, counting the 99,197 lines containing `function` in a ~380 MB corpus of real source. Reproduce with `tarn find -c` vs `rg -c` on any large file.

CONTENTS

What's inside

01 **Why agents fight their tools.** 3

02 **The surface: one binary, 25 commands.** 3

03 **Speed: at parity with ripgrep, zero deps.** 4

04 **How — mmap, all cores, SIMD, no crates.** 5

05 **An honest negative result: the index.** 6

06 **The evidence: what ~57,000 calls say.** 7

07 **Trust is the product.** 8

HOW TO READ THIS

Every claim is cashable.

The performance figures are medians of real runs on a real corpus against real tools (ripgrep 14, ugrep 7.5). Where a thing didn't work, it's in here too — section 05 is a feature we built, measured, and deleted. The tool is on GitHub; `tarn help --json` emits its whole surface for an agent to learn in one call.

01

THE PROBLEM

Why agents fight their tools.

A coding agent drives tools built for humans: open a file, scroll, count lines, hope the line numbers didn't move since the last read. The friction is real and it compounds across a session.

CONTEXT BURNED

whole files

read into the prompt just to find one symbol. grep gives lines without structure; cat gives everything without shape.

LINE DRIFT

off-by-N

an edit keyed to a line number that moved between read and write clobbers the wrong line. No guard, no recovery.

NO MACHINE SURFACE

prose

human tools answer in prose an agent must re-parse. No stable JSON, no exit-code contract to branch on.

02

THE SURFACE

One binary, 25 commands.

Structural, surgical, and self-describing. `tarn help --json` returns the whole manifest — every command, usage, example, and exit code — so an agent learns the tool in a single call.

NAVIGATE (NO LSP)

tree · outline · defs · refs · peek · find

Go-to-definition, find-references, repo map, and grep that hands you the enclosing definition of every hit — from heuristics, no language server.

EDIT BY MEANING

replace · insert · delete · apply · patch · batch · rename

`--def <name>` edits a whole function. `--expect` refuses if the target moved. `git diff | tarn patch` applies a unified diff, relocating drifted hunks.

CONFIG & VERIFY

json · toml · yaml · check · diff

Get/set/delete by path, format-preserving (comments & key order kept). Hygiene gate and a fast line-numbered diff. Exit codes: 0/1/2/3.

THE CONTRACT

Deterministic, guarded, chainable.

Every edit preserves line endings and never reflows untouched lines; `--dry-run` previews; multi-file `apply`/`patch` are atomic with rollback; `--json` on read & edit commands lets one command feed the next. An agent can act without a defensive re-read.

03

THE SPEED

At parity with ripgrep. Zero dependencies.

ripgrep is the fastest search tool in wide use — SIMD, mmap, a tuned parallel walk. tarn matches it on a directory and *beats* it on a single large file, with not one crate dependency.

COUNT MATCHES IN ONE 380 MB FILE — LOWER IS FASTER

30,720 matching lines. Warm cache, median of 15 runs (figures bounce ±~15% run to run).

tarn is ~1.3× faster than ripgrep here — and ~45× faster than the system grep (ugrep 7.5: ≈ **2.2 s**, off this chart).

COUNT MATCHES ACROSS 3,000 FILES (~380 MB) — LOWER IS FASTER

The recursive case. Warm cache, median of 15 runs.

Parity — ripgrep edges it ~1.1×. Both fan files out across cores; ripgrep's tuned per-file machinery still leads slightly.

THE HONEST CEILING

You can't count faster than you can read.

tarn counts a 380 MB file in ~42 ms; merely `cat`-ing it takes ~36 ms. tarn is at the memory-bandwidth wall — it counts in about the time it takes to read the bytes at all. So "10× faster" on a single scan isn't an optimization, it's physics. (The 10× hides elsewhere — see section 05.)

04

THE HOW

Fast with no crates.

ripgrep's raw byte-scan beats anything you'd hand-write in std. tarn wins the difference back three other ways — all standard library, no dependencies.

NO COPY

mmap (libc FFI)

Map the file and scan straight from the page cache — no `fs::read` copy of hundreds of MB. Same FFI tarn already uses for `stty`.

ALL CORES

std::thread

ripgrep searches one file on one thread; tarn splits it. `\n` is always a UTF-8 boundary, so each chunk counts independently and the sum is exact.

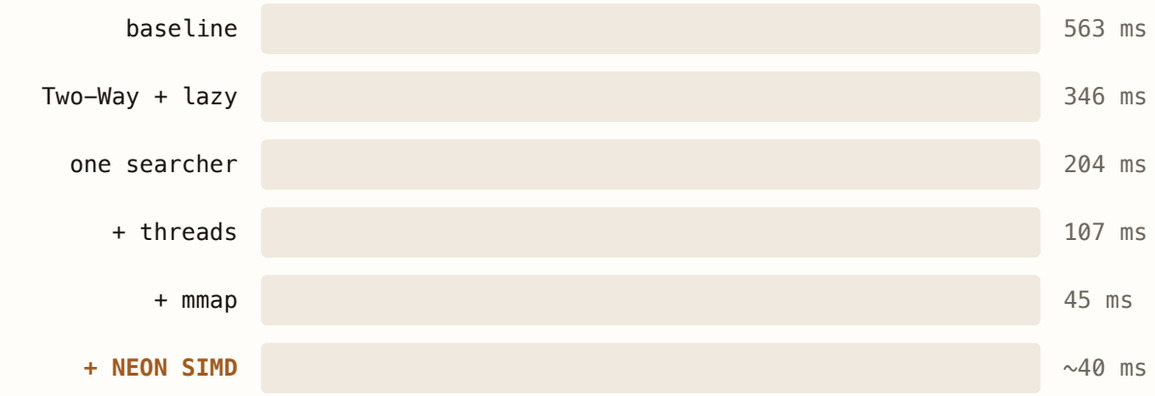
SIMD, STD

NEON via core::arch

A 16-byte-at-a-time `memchr` in NEON intrinsics — which live in `core::arch`, the standard library, not a crate. AddressSanitizer-clean.

THE JOURNEY — TARN FIND -C ON THE SAME 380 MB FILE

Each step is a std-only change. Lower is faster.



563 → ~40 ms: a ~14× speedup, entirely within the standard library.

BONUS: THE DIFF RENDERER

~7 s → ~26 ms on a 40k-line file.

A full-matrix LCS was a latent O(n·m) memory bomb. Trimming the common prefix/suffix and running the quadratic part only on the differing middle made a one-line change in a 40,000-line file diff in ~26 ms instead of ~7 s — backing every `--diff` flag. And `tarn batch` runs a whole command session in one process: ~10,500 edits/sec (~34× over per-call spawn), since an agent's edit loop is bottlenecked by OS process spawn, not by tarn.

05

THE CARDINAL RULE

Measure it. Then maybe delete it.

A tool earns trust by what it refuses to ship. We built a persistent navigation index to make repeated `defs`/`refs` sublinear — then measured it and pulled it.

THE PREMISE

91%

of a `defs` query on a 1,688-file repo is re-reading and re-parsing files that didn't change (~142 of ~156 ms). Cache that, the theory went, and queries go ~10×.

THE MEASUREMENT

1.4×

what the built, correct, never-stale per-file cache actually delivered. Not 10×.

THE REASON

decode

each `tarn` call is a fresh process that re-reads and re-deserializes the whole index. The floor is decode + stat, not the parse we cached.

WHAT SHIPPED INSTEAD: NOTHING

1.4× doesn't justify persistent state.

A new on-disk format and a new failure surface, for 1.4×, on a tool whose whole appeal is "just point it at files" — not worth it. The prototype was reverted; the analysis is recorded in `docs/INDEX.md`. A real 10× would need a resident daemon or an mmap'd zero-decode index — a much larger change, parked until the latency is actually felt. **The honest negative result is part of the spec.**

06

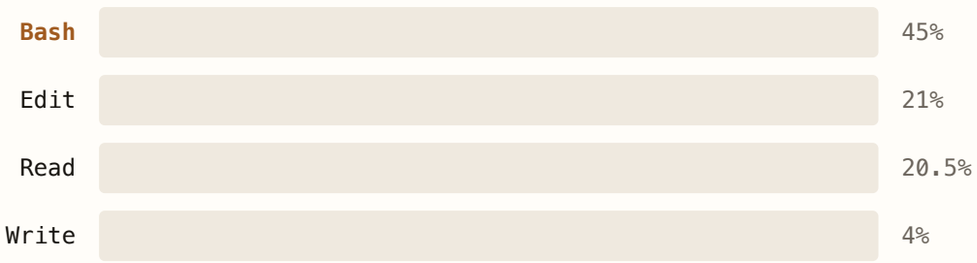
THE EVIDENCE

We stopped guessing. We counted.

Every feature so far came from a hunch about how an agent works a repo. Good hunches — but hunches. So we pulled **1,173 real Claude Code session transcripts (a June 2026 snapshot)** and counted what the agents in them actually did: **~57,000 tool calls**, tallied by hand, no sampling. Two findings, two features — neither was on the roadmap before we looked.

WHAT ~57,000 AGENT TOOL CALLS ACTUALLY WERE

1,173 Claude Code sessions (June 2026 snapshot), every call tallied. ~90% are file/shell ops — tarn's domain.



The rest is thinking and talking. The work an agent does to *code* is overwhelmingly running commands and changing files.

INSIDE BASH

34%

grep, by a mile.

`grep` was ~15,800 calls — a third of all CLI-utility use, ~100× more than ripgrep. So tarn's search speaks it: `tarn find -e/--regex`, a built-in regex engine, zero crates.

THE ENGINE

linear

Thompson/Pike NFA.

Regex the safe way — no catastrophic backtracking. `(a*)*b` on a 50,000-char line returns in 0.00s. Cross-validated against `grep -E`: no unexplained mismatches.

INSIDE EDIT

81%

Edits span lines.

9,661 of 11,872 edits touched more than one line. So `tarn replace <file> A-B <text>` swaps a whole line range for multi-line text — same `--expect` / `--diff` / `--dry-run`.

WHAT IT ISN'T

A useful subset, named as one.

The regex engine is a grep-ish subset, not PCRE — no backreferences, lookaround, or `\b`. Unsupported syntax errors *loudly* rather than matching the wrong thing, and literal substring stays the default. Structural `outline`/`defs` is still heuristic, with multi-line method signatures a known, deferred item. Built from the data, bounded by the truth — zero crates, 150 tests, every claim cashable.

07

THE PRODUCT

Trust is the product.

Speed gets a tool tried; trust gets it adopted. A search tool that returns one wrong-but-plausible answer gets pulled from the harness forever.

Adversarial review on every feature. A ruthless reviewer agent gated each change before it shipped — and earned its keep: it caught a real `.env` data-loss bug (a non-UTF-8 byte made `set` wipe the file), a CRLF counting seam, and exit-code contract violations, each fixed at the root.

Counts proven exact. The SIMD scanner is differential-tested against a scalar oracle — 900+ fuzz cases plus a 3,000-input run, zero mismatches — and its counts match `rg/grep` on the benchmark corpus. The unsafe NEON code is AddressSanitizer-clean.

Edits never corrupt. Line endings (LF/CRLF) and final-newline state are preserved on every edit; format-preserving config edits keep comments and key order; multi-file batches roll back as a unit.

150 tests, clippy & fmt clean, CI green. Every benchmark in this paper is reproducible from the repo. Where a number depended on the test machine, it's labelled. Nothing here is aspirational.

DEFINITION OF DONE

The tool an agent would choose.

Fast enough that you never reach for grep. Structural, so you jump to a symbol instead of scrolling. Guarded, so an edit can't silently land in the wrong place. Self-describing, so any agent learns it in one `tarn help --json`. And honest enough to delete the feature that didn't earn its place. Built by an agent — for the tools it wished it had.

tarn.

THE EDITOR YOUR AGENT WISHES IT HAD · BY AN AGENT, FOR AGENTS